$\boxed{17}$ $r: \ x = y = z - a$

$s: \ \frac{2x-1}{3} = \frac{y+3}{-2} = \frac{z-2}{0}.$

- a elkar ebakitzeko
- ebakitze puntua.

$s: \ \frac{x - 1/2}{3/2} = \frac{y+3}{-2} = \frac{z-2}{0}$

$\bar{d}_r = (1, 1, 1)$ $\qquad \bar{d}_s = (3/2, -2, 0)$

$P_r = (0, 0, a)$ $\qquad P_s = (1/2, -3, 2)$.

- a ELKAR EBAKITZEKO.

$\boxed{\overline{P_rP_s}\,(\bar{d}_r \wedge \bar{d}_s) = 0.}$ $\quad \boxed{|M'| = 0}$

$\overline{P_rP_s} = \overline{OP_s} - \overline{OP_r} = (1/2, 3, 2) - (0, 0, a) = (1/2, 3, 2-a)$

$$\begin{vmatrix} 1 & 3/2 & 1/2 \\ 1 & -2 & 3 \\ 1 & 0 & 2-a \end{vmatrix} = 0$$

$-2(2-a) + \frac{9}{2} + 1 - \frac{3}{2}(2-a) = 0$

$-4 + 2a + \frac{9}{2} + 1 - \frac{6}{2} + \frac{3a}{2} = 0$

$\frac{7a - 21}{2} = 0 \rightarrow \boxed{a = 3}$

- Ebakitze puntua aurkitzeko parametrikoetara jotzen dugu → parametro desberdinapot

$r: \begin{cases} x = 0 + \lambda \\ y = 0 + \lambda \\ z = 3 + \lambda \end{cases}$ $\qquad s: \begin{cases} x = 1/2 + 3/2\,\mu \\ y = -3 - 2\mu \\ z = 2 \end{cases}$

$x \rightarrow \begin{cases} \lambda = 1/2 + 3/2\,\mu \\ \lambda = -3 - 2\mu \\ 3 + \lambda = 2 \end{cases} \Rightarrow \begin{cases} \lambda - 3/2\,\mu = 1/2 \\ \lambda + 2\mu = -3 \\ \lambda = -1 \end{cases}$

Sistema ebatzi! $\Longrightarrow$ Beraz ebakitze puntua:

$\boxed{\lambda = -1} \rightarrow \boxed{\mu = -1}$

$r: \rightarrow \begin{matrix} x = -1 \\ y = -1 \\ z = 2. \end{matrix}$ $\qquad \boxed{(-1, -1, 2)}$

$\boxed{18}$ $r: \begin{cases} x = 5+4\lambda \\ y = 3+\lambda \\ z = -\lambda \end{cases}$ $s: \frac{x}{m} = \frac{y-1}{3} = \frac{z+3}{n}$

kalkulatu m eta n, paraleloak izateko.

Paraleloak izateko $\bar{d}s$ eta $\bar{d}r$ proportzionalak izan behar dira

$\bar{d}r = (4, 1, -1)$ $\frac{4}{m} = \frac{1}{3} = \frac{-1}{n}$

$\bar{d}s(m, 3, n)$ $\boxed{12 = m}$ $\boxed{n = -3}$

kasu honetan paraleloak edo bat egin daitezke
Aztertu egiten da ea $P_r \notin s$.

$P_r(5,3,0) \longrightarrow s: \frac{5}{12} = \frac{3-1}{3} = \frac{0+3}{-3}$

$\frac{5}{12} \neq \frac{2}{3} \neq -1$

Beraz $P_r \notin s \rightarrow$ $\boxed{\text{PARALELOAK dira}}$

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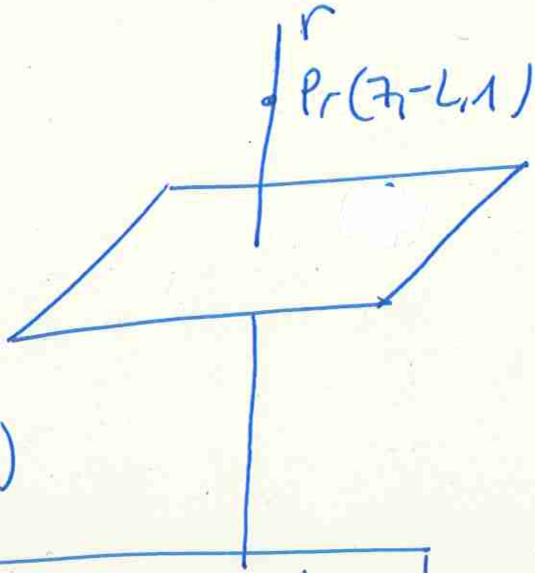
$$\left.\begin{array}{l} \pi: x+2y=5 \\ r \perp \pi \\ P_r(7,-2,1) \in r \end{array}\right\}$$



r zuzeuoren $\bar{d}_r = \bar{n}_\pi = (1,2,0)$

$$r: \begin{cases} \bar{d}_r\,(1,2,0) \\ P_r\,(7,-2,1) \end{cases} \Rightarrow \boxed{r: \begin{cases} x = 7+\lambda \\ y = -2+2\lambda \\ z = 1 \end{cases}}$$

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$P_s(-1,5,0)$

$$s \parallel r: \begin{cases} x+y=1 \\ y-z=2 \end{cases}$$

$$s: \begin{cases} \bar{d}_s = \bar{d}_r = \bar{n}_1 \wedge \bar{n}_2 = (1,1,0) \wedge (0,1,-1) \\ P_s(-1,5,0) \end{cases}$$

$$\bar{d}_r = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 0 \\ 0 & 1 & -1 \end{vmatrix} = -\hat{i} + \hat{k} + \hat{j} = (-1,1,1)$$

$$\boxed{s: \begin{cases} x = -1-\lambda \\ y = 5+\lambda \\ z = 0+\lambda \end{cases}}$$